Supplemental Online Content

Mayor N, Light A, Gopalakrishnan A, et al. Prostate Imaging Standards for Screening Magnetic Resonance Imaging (PRISM): international consensus recommendations. *JAMA Oncol*. Published online June 11, 2026. doi:10.1001/jamaoncol.2026.1711

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eFigure 8. Forest Plots Showing Pooled Estimates of the Ratio of Clinically Significant (CS) to Clinically Insignificant (CI) Prostate Cancers for the Primary Threshold Employed in Each Study (Top) and for the PI-RADS ≥3 (Middle) and PI-RADS ≥4 (Bottom) Subgroups

This supplementary material has been provided by the authors to give readers additional information about their work.

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eTable 1. MRI Thresholds for Biopsy Recommendation

Study	Primary threshold	Other reported thresholds
IP1-PROSTAGRAM	PI-RADS ≥3	PI-RADS ≥3, Likert ≥3, Likert ≥4
MVP Pilot	Biopsy for all	PI-RADS ≥3, PI-RADS ≥4
MVP	PI-RADS ≥4	Nil
PROSA	PI-RADS ≥3	Nil
Re-IMAGINE	Stage-gated reporting	Nil
VISIONING	PI-RADS ≥3	Nil

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eTable 2. Participant Experience With Screening MRI

Participant	Country	Specialty	Experience
Aiman Haider	United Kingdom	Pathology	8 years reporting MRI-targeted biopsy, 1000 biopsies reported/year.
Amish Lakhani	United Kingdom	Radiology	10 years experience reporting MRI, 1000/year, 100-150 non-contrast
Andrew Smith	United Kingdom	Pathology	5 years reporting MRI-targeted biopsy, 350 biopsies reported/year.
Anwar Padhani	United Kingdom	Radiology	500 diagnostic MRIs reported/year
Caroline Moore	United Kingdom	Urology	20 years experience with MRI in routine clinical practice
Francesco Giganti	United Kingdom	Radiology	15 years reporting MRI, 2,000 MRIs reported/year, 300 non-contrast
Hashim Ahmed	United Kingdom	Urology	15 years experience with MRI in routine clinical practice
Henry Tam	United Kingdom	Radiology	11 years reporting MRI, 400/year, 100 non-contrast
Ivo Schoots	Netherlands	Radiology	15 years reporting MRI in routine clinical practice
Mark Emberton	United Kingdom	Urology	20 years experience with MRI in routine clinical practice
Masoom Haider	Canada	Radiology	20 years reporting MRI, 400/year, 50 non-contrast
Massimo Valerio	Switzerland	Urology	16 years experience with MRI in routine clinical practice
Ola Bratt	Sweden	Urology	10 years experience with MRI in routine clinical practice
Oliver Hulson	United Kingdom	Radiology	10 years reporting MRI, 350/year, 150 non-contrast
Sadhna Verma	United States of America	Radiology	24 years reporting MRI, 1000-1200/year, 100-120 non-contrast
Samuel Withey	United Kingdom	Radiology	7 years reporting MRI, 1500/year, 300 non-contrast
Shonit Punwani	United Kingdom	Radiology	23 years reporting MRI, 2000/year, 300 non-contrast
Taimur Shah	United Kingdom	Urology	10 years experience with MRI in routine clinical practice
Tim Dudderidge	United Kingdom	Urology	16 years experience with MRI in routine clinical practice
Tobias Nordstrom	Sweden	Urology	Not reported
Tristan Barrett	United Kingdom	Radiology	16 years reporting MRI, 750/year, 300 non-contrast

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eTable 3. Core Assumptions Upon Which Recommendations Were Made

Assumption
Screening refers to an investigation in an apparently healthy person who may be at increased risk of a disease or condition, enabling earlier treatment or better-informed decisions.
Screening MRI refers only to an MRI in a man with no prior history of prostate adenocarcinoma
Unless stated otherwise, screening MRI refers to MRI in men who have not been risk-stratified prior to being invited to screening
Unless otherwise stated, men undergoing screening MRI are fit with no treatment-limiting comorbidities
All MRI techniques are readily available
An MRI sequence, where acquired, is of adequate quality to be interpreted with diagnostic intent. This includes no or minimal artefact
Participants undergoing MRI have no contraindication to any MRI scan type, including contrast allergy
All diagnostic procedures and treatments (including expertise in interpretation and application) are readily available

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eTable 4. Characteristics of Studies Included in the Systematic Review

Study acronym	First author	Year of publication	Recruitment method	Study design	Inclusion Criteria	Exclusion Criteria
IP1-PROSTAGRAM	Eldred-Evans	2019	Dedicated direct-to-community recruitment strategy	Prospective, blinded, population-based screening study	• Age 50-69 years • Life expectancy (LE) ≥10 years	• PSA or prostate MRI in previous 2 years • Urinary infection or prostatitis in previous 6 months • History of prostate biopsy, prostate cancer, or any contraindication to MRI
MVP Pilot	Nam	2016	News advertisement in large local newspaper	Single-arm prospective cohort study (pilot)	• Age 50-69 years • English speakers	• Family history of prostate cancer • History of prostate biopsy • History of any cancer other than non-melanomatous cutaneous cancer
MVP	Nam	2022	Newspaper and radio advertisements	Phase III open-label randomised controlled trial	• Age ≥50 years • LE ≥10 years	• History of any prostate biopsy • PSA within 3 years • urinary difficulty symptoms • History of PCa in ≥1 first degree relatives diagnosed <50 years • International Prostate Symptom Score (IPSS) ≥8 • 5-alpha reductase inhibitor medications • Contraindication to MRI • Abnormal DRE
PROSA	Messina	2023	Employees of host institution (large tertiary hospital)	Prospective, randomized, single-center	• Age 49-69 years (40-69 if family history PCa) • LE ≥10 years	• Contraindication to MRI or prostate biopsy • History of PCa or prostate biopsy, • Active urinary tract infection or prostatitis)
Re-IMAGINE	Moore	2023	Invitation by screening patient databases at eight London GP surgeries	Single-arm prospective cohort feasibility study	• Age 50–75 years	• History of PCa
VISIONING	Wetterauer	2024	Through general practitioners, neighbouring clinics, and urologists in private practice referring interested men.	Single-centre prospective cohort study	• Age ≥50 years (≥45 years if African ancestry or family history PCa) • LE ≥10 years	• Prior prostate biopsies • Acute urinary tract infection • Chronic Prostatitis Symptom Index score ≥19 • IPSS ≥20 • Contraindication to MRI

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eTable 5. Characteristics of Patients in Each Study Included in the Systematic Review

Study acronym	Participants, n	Upfront MRI, n	Age (yr) Mean ± SD/median (IQR)*	PSA (ng/ml)—mean ± SD/median (IQR)*	Family history	Abnormal DRE	Black ethnicity
IP1-PROSTAGRAM	411	406	57 (53–61)	0.91 (0.56–1.74)	43/403 (10.7%)	18/405 (4.4%)	132/408 (32.4%)
MVP Pilot	50	50	61 (55–68)	3.03 (1.90–5.47)	NR	5/47 (10.6%)	NR
MVP	525	246	67.7±7.3	NR	NR	0/525 (0.0%)	5/504 (1.0%)
PROSA	351	175	58.8 (46–69) [median, range]	2.63 (0.21–95.4) [median, range]	89/351 (25.4%)	NR	NR
Re-IMAGINE	309	303	61.9±7.2	1.2 (0.7–2.2)	41/308 (13.3%)	NR	15/374 (4.0%)
VISIONING	269	246	58 (53–64)	1.26 (0.72–2.84)	38/229(16.6%)	8/229 (3.5%)	NR

*Of those undergoing upfront screening MRI

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eTable 6. MRI Specifications, Protocols, Indications for Biopsy, and Biopsy Protocols

Study acronym	Field strength	MRI type and coils	Protocol and duration	Duration	Patient preparation	Interpretation	Indications for biopsy	Biopsy protocol
IP1-PROSTAGRAM	3T Siemens Magnetom Verio: 338 (82.8%) 1.5T Siemens Magnetom Aera: 70 (17.2%)	bpMRI with pelvic phased-array coils	T2-W axial & sagittal TSE, DWI, ADC (b values 0, 150, 400, 1000), high b value (1500)	~15 min	IV buscopan 20mg	Single expert radiologist (20% double reported)	PI-RADS >/=3	Transperineal (TP), systematic 12-core and image fusion-targeted biopsy
MVP Pilot	3T Achieva MRI system (Philips Healthcare, Best, The Netherlands	mpMRI with 6-channel SENSE cardiac surface coil (Philips Healthcare) over pelvis	Multiplanar T2W-I, DWI, ADC, (b values 100, 400, 1000), DCE	NR	40 mg butylscopalamine bromide (Buscopan®) IM	Single expert radiologist	Biopsy offered to all	Transrectal ultrasound (TRUS) guided random 12-core and cognitive targeted biopsy (4 cores per target)
MVP	3T Siemens Prisma	bpMRI with 6-channel phased array	T2-W axial, coronal & sagittal, DWI, ADC (b values 50,200,400,600,1000), high b value (1600)	30 min	NR	Single expert radiologist	PI-RADS >/=4	TRUS-guided systematic 12-core and image-fusion targeted biopsy (4 cores per target)
PROSA	3T GE Discovery 750 and MAGNETOM Vida Siemens	bpMRI with 32-channel surface phased-array pelvic coil	T2-W axial and coronal FRFSE, DWI, ADC b values 100, 800, 1000, high b value (2000)	15 ± 1.3 min	Evacuative enema 1 h pre-MRI Low-fiber diet 3 days pre-screening.	Two expert radiologists (decision by most-experienced if disagreement)	PI-RADS >/=3	TRUS-guided image-fusion targeted biopsy
Re-IMAGINE	3T	bpMRI	T2-W axial TSE, DWI, high b value (2000), no ADC	<10 min	NR	Two radiologists (third if disagreement)	Novel "stage-gated" reporting system	TP systematic and image fusion-targeted biopsy
VISIONING	3T MAGNETOM Prisma; Siemens	bpMRI	T2-W coronal and sagittal (re-sliced from transverse) TSE, DWI, ADC b values (0, 800)	NR	NR	Two radiologists	PSA >10 PSAD > 0.15 Abnormal DRE Phase I: PI-RADS >/=3 Phase II: PI-RADS >/=4 or PI-RADS 3 + PI-RADS >/=3 on repeat MRI at 6mo	Systematic 18-core biopsy and: Phase 1: TRUS-guided image-fusion targeted biopsy Phase 2: Robotic-assisted TP image-fusion targeted biopsy

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eTable 7. Summary of Outcomes and Estimates of Heterogeneity for Meta-Analysis at PI-RADS ≥3 Threshold for MRI Positivity

Outcome	Pooled outcome	95% CI	Q	df	p_Q	I2 (%)	95% CI (%)
Biopsy recommendation	27.2%	10.5-44%	49.2	3	<0.001	97.2	88.8-96.3
Biopsy adherence	84.1%	58.4-109.8%	NA	2	NA	NA	NA
GG≥2 detection	4.5%	1-8%	9.0	2	0.01	84.2	19.6-92.1
GG1 detection	1.8%	0.9-2.7%	2.3	2	0.31	0.8	0-84.2
PPV (GG≥2)	25.2%	18.3-32.1%	1.6	2	0.45	0	0-81.4
GG≥2:GG1 ratio	2.29	1.29-4.04	0.2	2	0.9	0	0-60

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eTable 8. Summary of Outcomes and Estimates of Heterogeneity for Meta-Analysis at PI-RADS ≥4 or Stage-Gated Approach Threshold for MRI Positivity

Outcome	Pooled outcome	95% CI	Q	df	p_Q	I2 (%)	95% CI (%)
Biopsy recommendation	16%	7.3-24.7%	15.8	3	0.001	93.4	47-91.6
Biopsy adherence	89.2%	79.2-99.2%	5.6	2	0.06	67.4	0-89.7
GG≥2 detection	7%	1.8-12.2%	17.8	3	<0.001	91.5	55-92.3
GG1 detection	1.1%	0.4-1.7%	2.5	3	0.48	0	0-75.7
PPV (GG≥2)	40.3%	19.6-61%	25.0	3	<0.001	87.2	71.9-93.9
GG≥2:GG1 ratio	3.9	1.88-8.09	4.1	3	0.25	25.3	0-81.2

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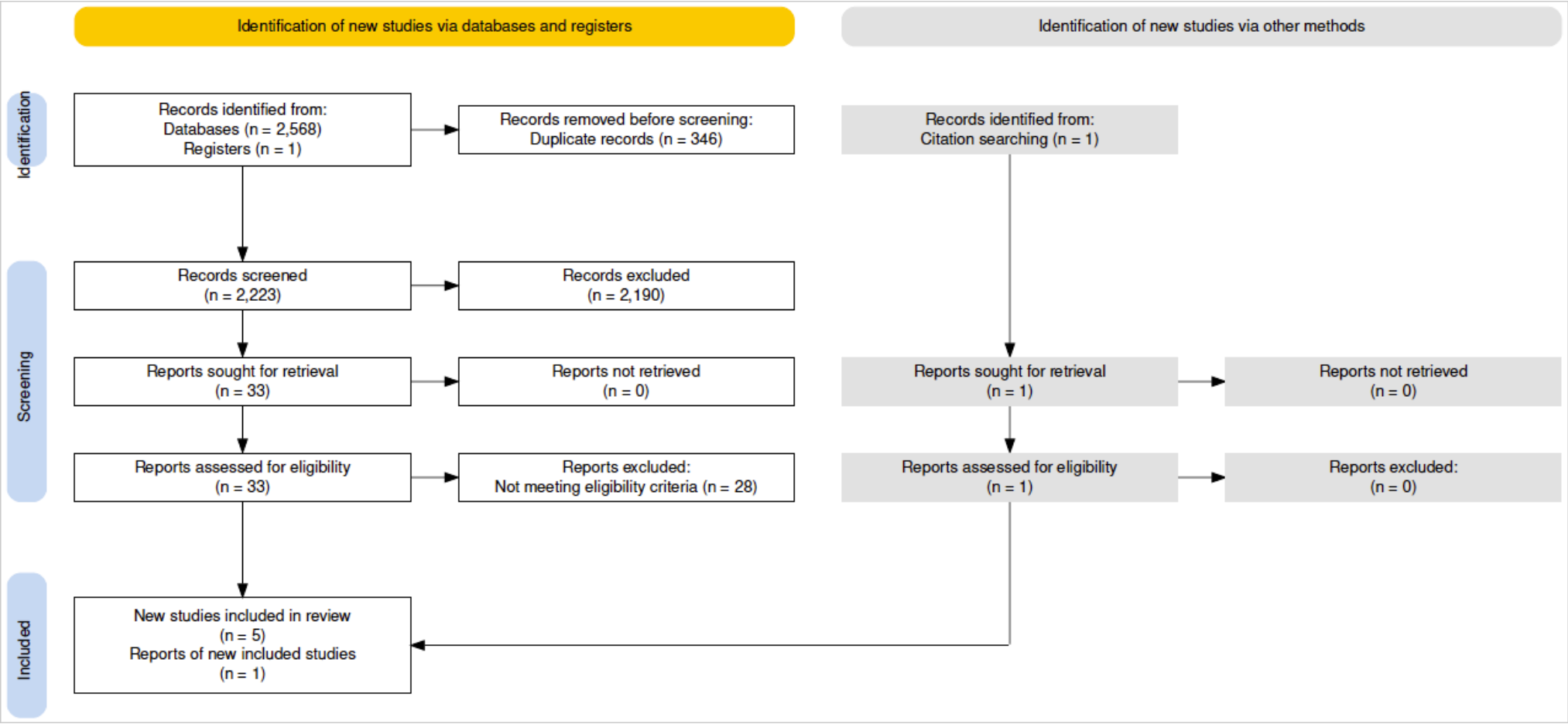
eTable 9. Diagnostic Criteria and Quality Rules for Stage 1 of the Stage-Gated Approach

Radiologists can only progress to Stage 2 (i.e. view the remaining planes and diffusion-weighted sequences of the abbreviated biparametric MRI) if there is a focal lesion on the axial high b, with a corresponding focal lesion/diffuse area on the axial T2W images.

Axial T2	Axial high b	Stage 1 outcome
Focal 4-5/5 or diffuse 3/5	Focal 4-5/5	Positive
Focal 4-5/5	Focal 1-2/5 or diffuse 3/5	Negative
Diffuse 3/5	1-2/5 or diffuse 3/5	Negative
1-2/5	ANY	Negative
ANY	1-2/5	Negative
Non-diagnostic (artefact)	1-2/5 or diffuse 3/5	Negative
1-2/5 or diffuse 3/5	Non-diagnostic (artefact)	Negative
4-5/5	Non-diagnostic (artefact)	Non-diagnostic
Non-diagnostic (artefact)	4-5/5	Non-diagnostic
Non-diagnostic (artefact)	Non-diagnostic (artefact)	Non-diagnostic

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eFigure 1. PRISMA Flow Diagram



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eFigure 2. Risk of Bias Assessment Using QUADAS-2 Tool

Study	Risk of bias: Patient selection				Risk of bias: Index test			Risk of bias: Reference standard			Risk of bias: Flow and timing					Concern regarding applicability		
	1	2	3	RoB1	1	2	RoB2	1	2	RoB3	1	2	3	4	Rob4	Patient selection	Index test	Reference standard
IP1-PROSTAGRAM	No	Yes	Yes	Unclear	Yes	Yes	Low	Yes	Unclear	Low	Unclear	No	No	Yes	High	Low	Low	Low
ReIMAGINE	Yes	Yes	Yes	Low	Yes	Yes	Low	Yes	Unclear	Low	Unclear	No	Yes	Yes	High	Low	Unclear	Low
MVP Pilot	No	Yes	Unclear	High	Yes	Yes	Low	Yes	Unclear	Low	Unclear	Yes	Yes	Yes	Low	Unclear	Low	Low
MVP	No	Yes	Yes	Unclear	Yes	Yes	Low	Yes	Unclear	Low	Unclear	No	No	Yes	High	Low	Low	Low
PROSA	Unclear	Yes	Yes	Unclear	Yes	Yes	Low	Yes	Unclear	Low	Unclear	No	No	Yes	High	Low	Low	Low
VISIONING	No	Yes	Yes	High	Yes	No	High	Yes	Unclear	Unclear	Unclear	No	No	Yes	High	Low	Low	Low

QUADAS-2 Signalling Questions

RISK OF BIAS

Patient Selection

1. Was a consecutive or random sample of patients enrolled?
2. Was a case-control design avoided?
3. Did the study avoid inappropriate exclusions?

Index Test

1. Were the index test results interpreted without knowledge of the reference standard?
2. If a threshold was used, was it pre-specified?

Reference Standard

1. Is the reference standard likely to correctly classify the target condition?
2. Were the reference standard results interpreted without knowledge of the index test?

Flow and Timing

1. Was there an appropriate interval between index test and reference standard?
2. Did all patients receive the same reference standard?
3. Were all patients included in the analysis?
4. Were all patients who received index test(s) also given the reference standard?

APPLICABILITY

Patient Selection

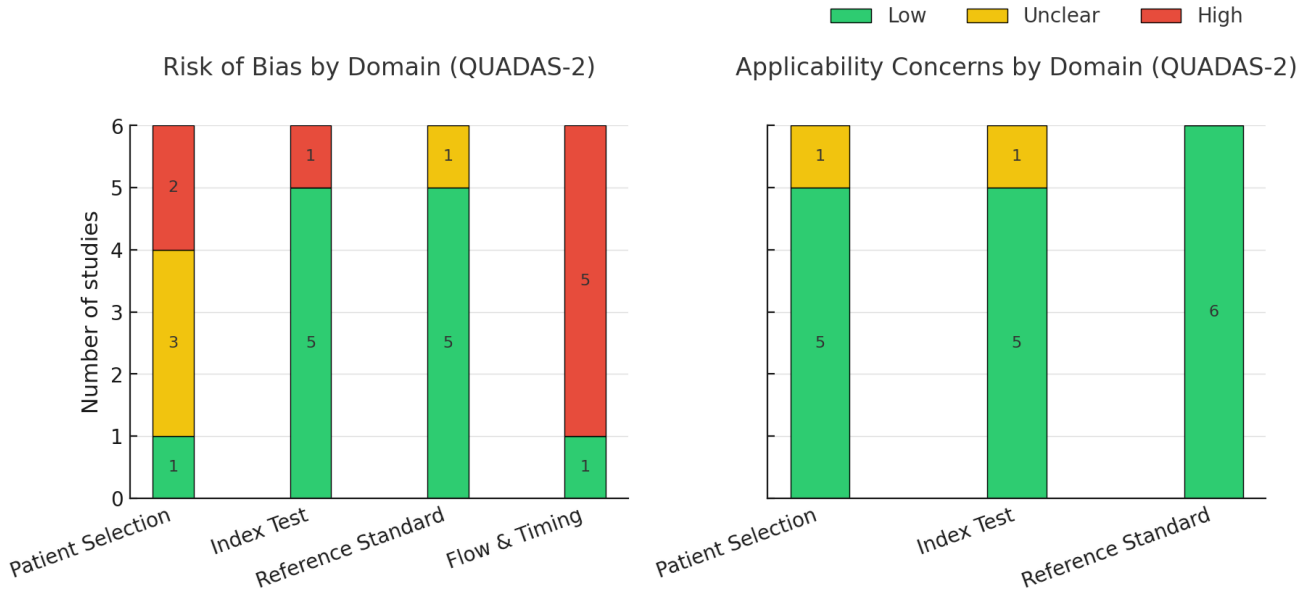
Are there concerns that the included patients and setting do not match the review question?

Index Test

Are there concerns that the index test, its conduct, or its interpretation differ from the review question?

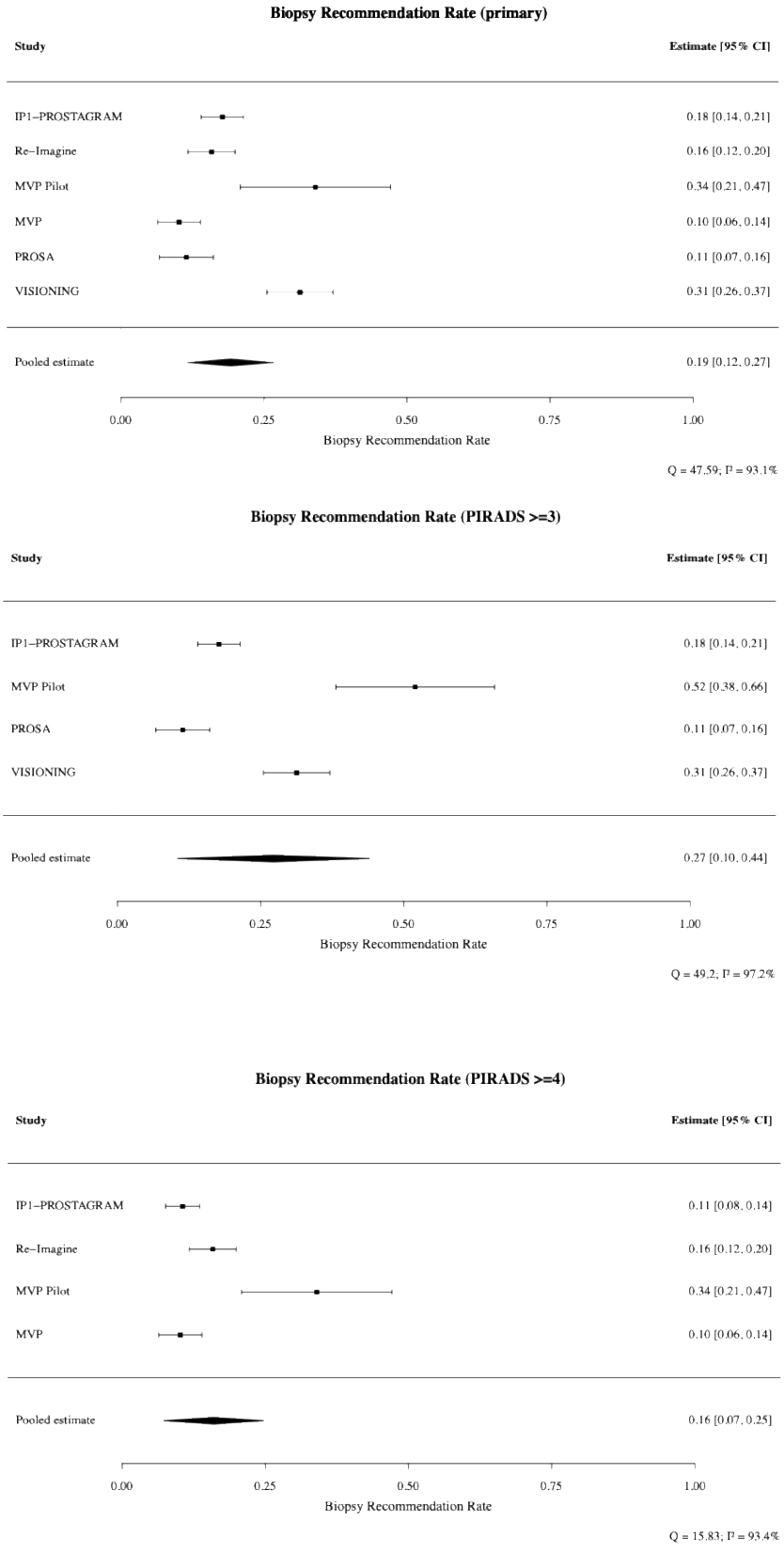
Flow and Timing

Are there concerns that the target condition as defined by the reference standard does not match the review question?



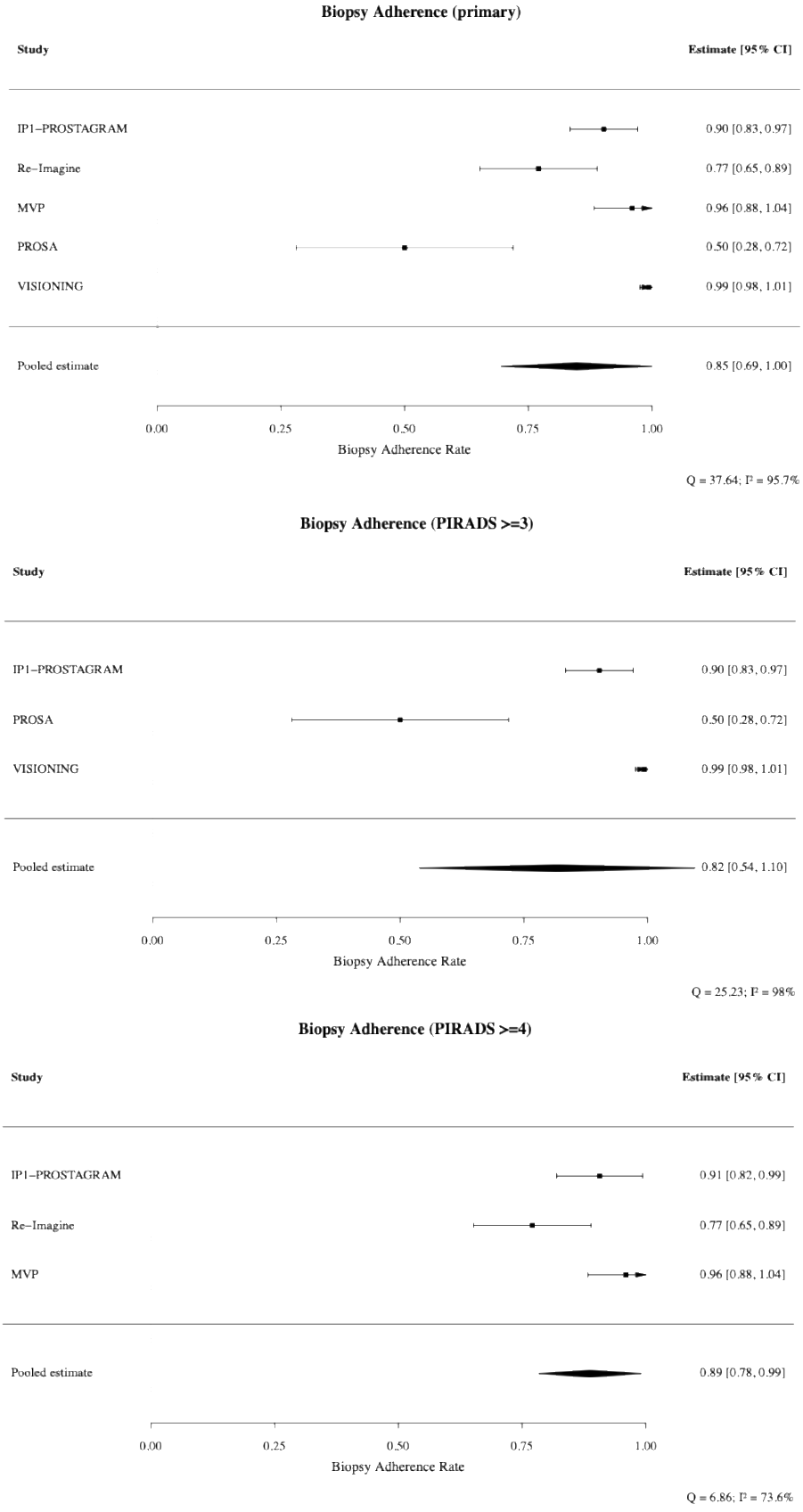
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eFigure 3. Forest Plots Showing Pooled Estimates of Biopsy Recommendation Rates for the Primary Threshold Employed in Each Study (Top) and for the PI-RADS ≥3 (Middle) and PI-RADS ≥4 (Bottom) Subgroups



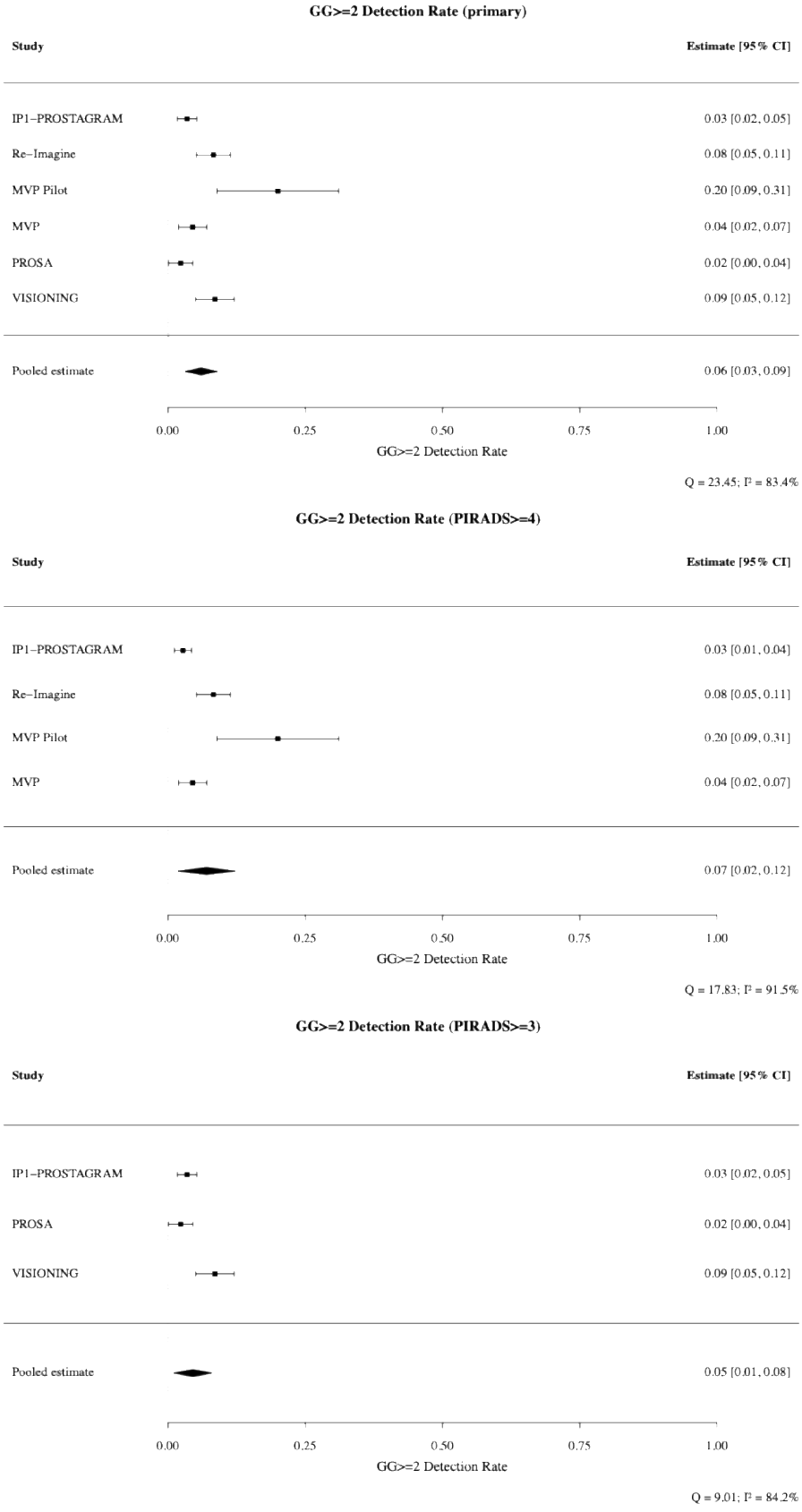
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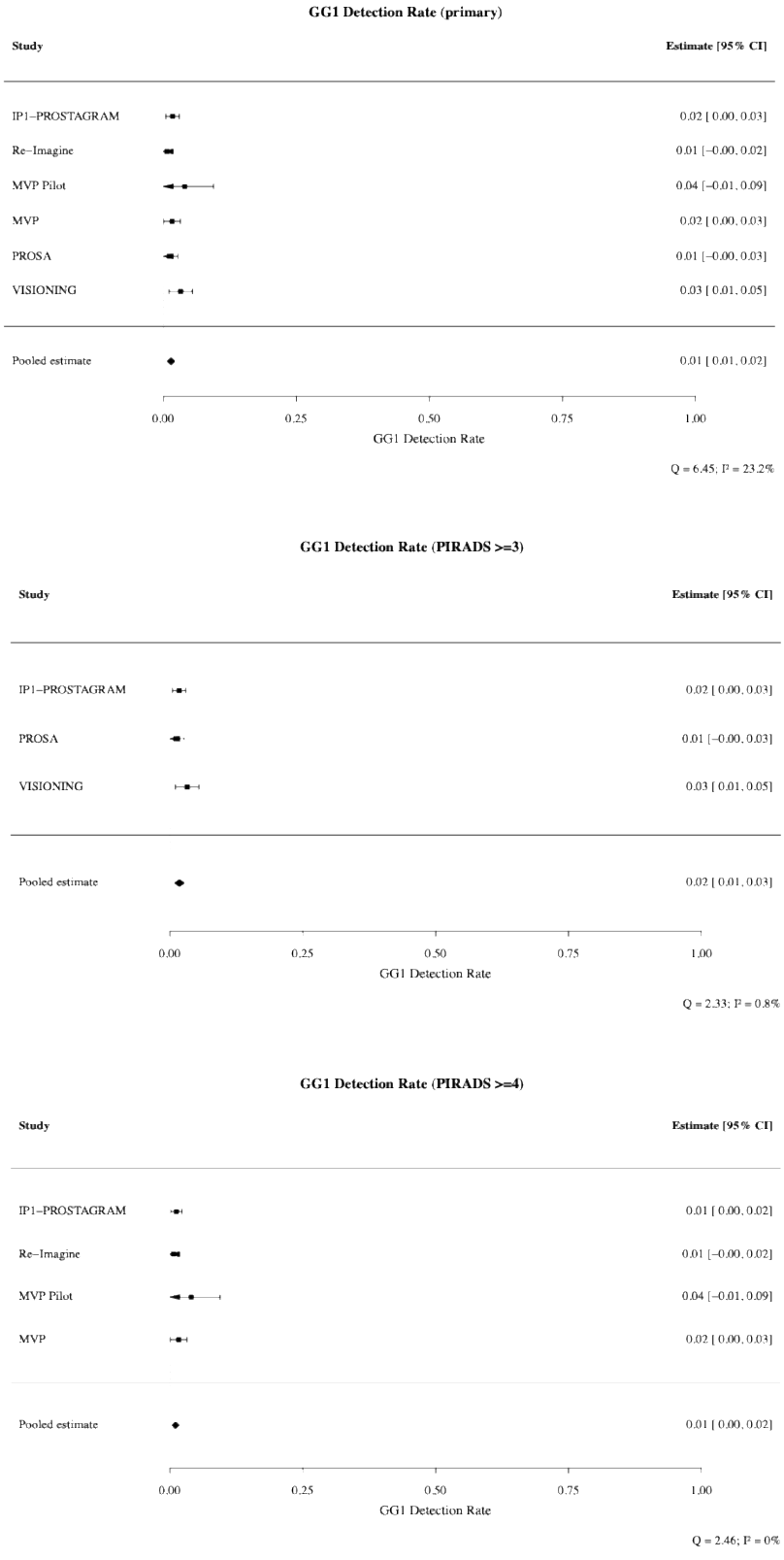
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eFigure 5. Forest Plots Showing Pooled Estimates of Clinically Significant Prostate Cancer (csPCa) Detection Rates for the Primary Threshold Employed in Each Study (Top) and for the PI-RADS ≥3 (Middle) and PI-RADS ≥4 (Bottom) Subgroups



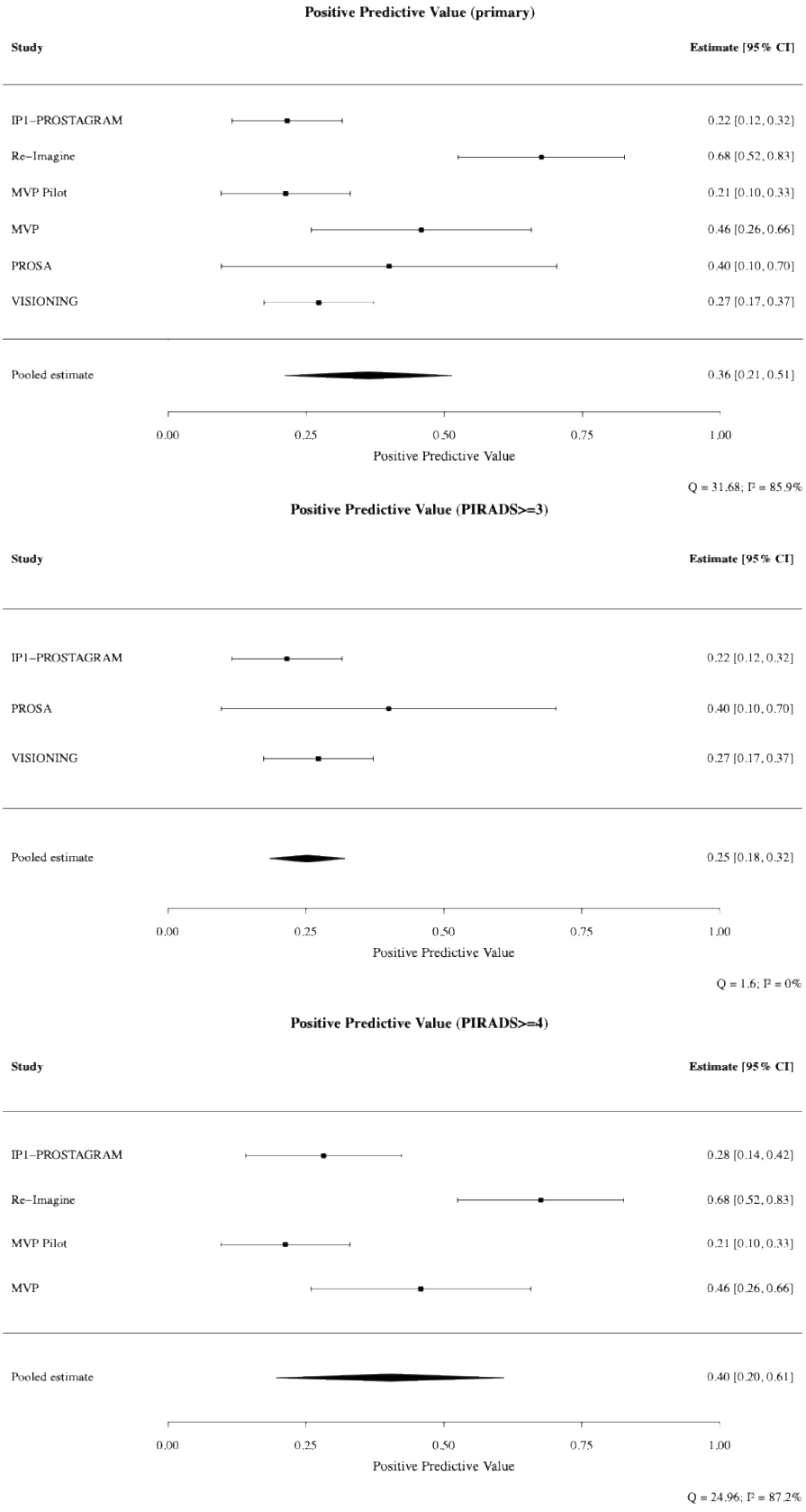
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eFigure 6. Forest Plots Showing Pooled Estimates of Clinically Insignificant Prostate Cancer (ciPCa) Detection Rates for the Primary Threshold Employed in Each Study (Top) and for the PI-RADS ≥3 (Middle) and PI-RADS ≥4 (Bottom) Subgroups



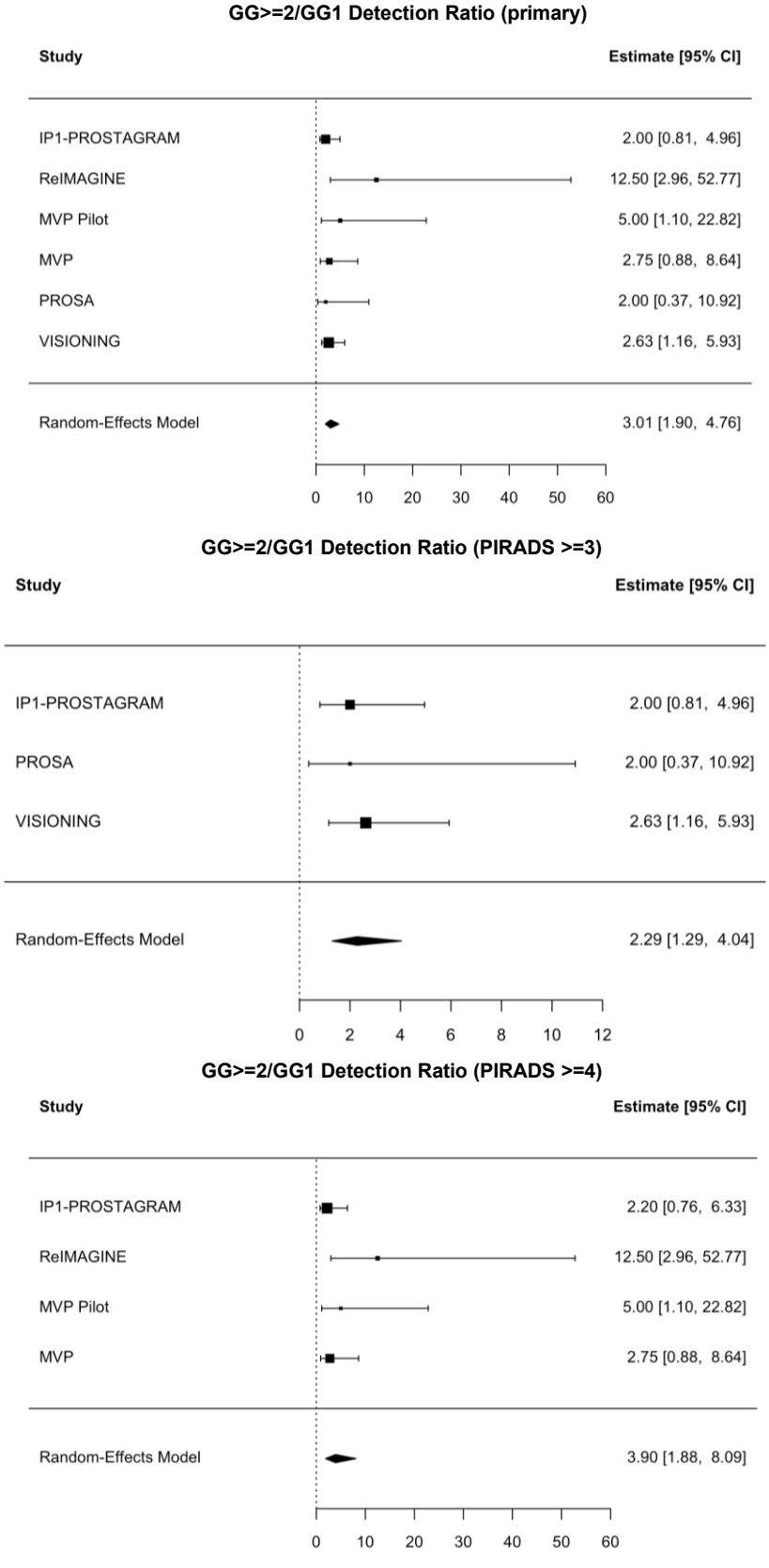
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eFigure 7. Forest Plots Showing Pooled Estimates of Positive Predictive Value (PPV) for Clinically Significant Prostate Cancer (GG ≥2) for the Primary Threshold Employed in Each Study (Top) and for the PI-RADS ≥3 (Middle) and PI-RADS ≥4 (Bottom) Subgroups



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